Homework 3: Financial Data Analysis

FMA4200@CUHK(SZ)

Due on March 28, 2026 (11:59pm)

Submit an electronic version of your answer to Blackboard (late submission will not be accepted).

1. Consider a MA(2) process:

$$X_t = \mu + \varepsilon_t + a_1\varepsilon_{t-1} + a_2\varepsilon_{t-2},$$

where ε_t is a white noise process with mean 0 and variance σ^2.

   (a) Is the model stationary?

   (b) Write down the invertibility condition of the model. Be as explicit as possible.

   (c) Find the mean, variance, autocovariance function and autocorrelation function of X_t.

   (d) Do you think the bitcoin returns can be modeled by a MA(q) model? Verify your answer with the data provided in the attached file.

2. Consider an AR(2) process:

$$X_t = c + b_1X_{t-1} + b_2X_{t-2} + \varepsilon_t,$$

where ε_t is a white noise process with mean 0 and variance σ^2.

   (a) Is the model invertible?

   (b) Write down the stationary condition of the AR(2) process. Be as explicit as possible.

   (c) Find the mean, variance, autocovariance function and autocorrelation function of X_t.

   (d) Do you think the bitcoin returns can be modeled by an AR(p) model? Verify your answer with the data provided in the attached file.